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GEOMETRIC IMAGE CLUSTERING AND REPRESENTATIVE SELECTION FOR HISTORICAL PHOTO COLLECTIONS

BACHELOR PROJECT REPORT

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ABSTRACT

This report addresses the challenge of organizing heterogeneous image collections from the Flickr Commons archive into geometrically consistent groups suitable for automated archival curation and downstream geolocalisation. Rather than targeting full 3D reconstruction, our primary objective is intelligent data pruning: given a coarse geographic cluster as input, we identify subgroups of images depicting the same landmark and select a single representative image per group for subsequent localisation against modern street-level imagery. Our pipeline combines CLIP-based visual pre-filtering with local feature matching (SIFT + RootSIFT with CLAHE preprocessing), dual RANSAC geometric verification, and LightGlue + SuperPoint symmetric verification. A weighted correspondence graph is built over each geo-cluster, and connected components define the final groups. A centrality-based criterion identifies the most geometrically connected image per group. On a dataset of **714** images across **164** geo-clusters, **101** of which are singletons by construction, our method processes the **63** non-trivial clusters and discovers **11** multi-image groups from the **32** clusters that yield at least one candidate pair after CLIP pre-filtering, with an average group size of **2.09** images, while reducing the pairwise comparison workload by up to **79%** through CLIP top- k pre-filtering on large clusters.

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CHAPTER 1

INTRODUCTION

Flickr Commons hosts millions of historical photographs contributed by cultural heritage institutions worldwide. These images, designated with no known copyright restrictions, represent an invaluable resource for digital humanities research. However, their heterogeneous nature (varying viewpoints, lighting conditions, image quality, and acquisition dates) makes large-scale computational analysis challenging.

This work is part of a broader research collaboration between the EPFL Digital Humanities Lab (DHLAB) and the Flickr Foundation, whose goal is to leverage social metadata to enhance AI-generated descriptions of archival images. Within this four-student project, our contribution focuses on the dataset preparation stage: given a collection of Flickr Commons images coarsely grouped by geographic proximity, we aim to discover subgroups of images depicting the same scene or landmark from overlapping viewpoints.

Unlike traditional structure-from-motion pipelines that require dense, high-quality image sets to produce complete 3D models, our goal is intelligent curation. We seek to collapse redundant photographs into a single stable representative, reducing the workload of the downstream localisation stage and filtering out images that share no geometric content with their neighbours. This grouping step serves a dual purpose: it reduces noise within each geo-cluster, and it identifies a single central image per group, namely the image with the most geometric connections to its neighbours, which serves as a candidate for localisation against Mapillary street-level imagery.

Our pipeline combines CLIP (Radford et al. 2021) visual pre-filtering with SIFT + RootSIFT feature matching and CLAHE contrast enhancement, dual RANSAC geometric verification (fundamental matrix and homography), and a LightGlue + SuperPoint (Lindenberger, Sarlin and Pollefeys 2023; DeTone, Malisiewicz and Rabinovich 2018) learned matcher. A correspondence graph is built over each geo-cluster, and connected components define the final groups.

CHAPTER 2

RELATED WORK

2.1 IMAGE RETRIEVAL FOR LARGE-SCALE COLLECTIONS

Scaling to large unordered photo collections requires an efficient retrieval front-end that narrows candidate pairs before expensive feature matching. Early approaches relied on bag-of-words representations (Sivic and Zisserman 2003); more recent work exploits compact global descriptors such as NetVLAD (Arandjelović, Gronát et al. 2016) or CLIP (Radford et al. 2021) embeddings, which transfer well to out-of-domain historical imagery without fine-tuning.

2.2 LOCAL FEATURE MATCHING AND GEOMETRIC VERIFICATION

SIFT (Lowe 2004) remains a competitive baseline for matching archival photographs due to its robustness to scale and rotation changes. The RootSIFT variant (Arandjelović and Zisserman 2012) improves matching accuracy by replacing the ℓ_2 distance with a Hellinger kernel. Geometric verification via RANSAC (Fischler and Bolles 1981) (estimating either the fundamental matrix or homography) is a standard post-processing step to discard false matches.

2.3 LEARNING-BASED MATCHERS

Learned matching networks such as SuperGlue (Sarlin et al. 2020) and LightGlue (Lindemberger, Sarlin and Pollefeys 2023) have recently outperformed hand-crafted methods on standard benchmarks. LightGlue in particular offers a favourable accuracy/speed trade-off and is well-suited as a high-precision verification stage applied after a cheaper SIFT pre-filter.

2.4 IMAGE CLUSTERING FOR HERITAGE COLLECTIONS

Prior work on photo tourism (Snavely, Seitz and Szeliski 2006) and landmark recognition (Chen et al. 2011) has demonstrated that community photo collections can be clustered into scene-consistent groups. Our work adapts these ideas to the specific challenges of Flickr Commons: mixed photographic quality, wide temporal range, coarse GPS metadata, and the primary goal of curation rather than reconstruction.

CHAPTER 3

METHOD

3.1 PROBLEM FORMULATION

Let $\mathcal{C} = \{I_1, I_2, \dots, I_n\}$ denote a geo-cluster of n images, produced by a preceding stage that groups photographs by geographic proximity using GPS coordinates. Our goal is to partition $\mathcal{C}$ into groups $\mathcal{G} = \{G_1, G_2, \dots, G_k\}$, where each group $G_j \subseteq \mathcal{C}$ contains images that share sufficient geometric overlap to be considered co-visible observations of the same scene. Images that share no overlap with any other image form singleton groups.

For each group G_j , we additionally identify a central image $I_j^* \in G_j$, defined as the image with the highest average geometric match score to its neighbours. This representative is passed to the downstream localisation stage, reducing the matching workload from all images in the group to a single stable candidate. The output assigns each image a group identifier `<geo_cluster_id>_<local_id>` and a flag `is_central`.

3.2 CANDIDATE RETRIEVAL

We exploit precomputed CLIP ViT-B/32 (Radford et al. 2021) embeddings: each image I_i is represented by a 512-dimensional L2-normalised vector $\mathbf{e}_i$, and the cosine similarity reduces to $s_{ij} = \mathbf{e}_i^\top \mathbf{e}_j$. The pipeline also supports DINOv2 (Oquab et al. 2024) as a fallback, but all reported experiments use the CLIP embeddings precomputed upstream. Candidate pairs are formed as:

$$\mathcal{P} = \{(i, j) \mid j \in \text{top-}k_i, s_{ij} \geq \tau, i < j\} \quad (3.1)$$

with $k = 10$ and $\tau = 0.75$, reducing comparisons from $\mathcal{O}(n^2)$ to $\mathcal{O}(nk)$. For our largest cluster of 67 images, this reduces $\binom{67}{2} = 2211$ pairs to approximately 467, a 79% reduction.

3.3 PREPROCESSING

Each image is resized to a maximum of 1400 pixels on its largest side. A border mask blanks the outermost 7% of pixels on each side to suppress institutional watermarks. Contrast is then enhanced with CLAHE (clip limit 2.0, tile grid 8×8 px).

3.4 LOCAL FEATURE MATCHING

For each pair $(I_i, I_j) \in \mathcal{P}$, we extract up to 4000 SIFT keypoints (Lowe 2004) and apply the RootSIFT transformation (Arandjelović and Zisserman 2012). A FLANN-based matcher with ratio test $r = 0.75$ establishes correspondences; pairs with fewer than 30 good matches are discarded.

3.5 GEOMETRIC VERIFICATION

STAGE 1: DUAL RANSAC

We fit both a fundamental matrix $\mathbf{F}$ (threshold 3.0px) and a homography $\mathbf{H}$ (threshold 4.0px) via RANSAC (Fischler and Bolles 1981), keeping the model with more inliers. Acceptance criteria:

$$n_{\text{good}} \geq 30 \quad (3.2)$$

$$n_{\text{inliers}} \geq 20 \quad (3.3)$$

$$r_{\text{inliers}} \geq 0.35 \quad (3.4)$$

STAGE 2: LIGHTGLUE VERIFICATION

Valid pairs are re-verified using LightGlue with SuperPoint keypoints (Lindenberger, Sarlin and Pollefeys 2023; DeTone, Malisiewicz and Rabinovich 2018) in the reverse direction (I_j, I_i) as a symmetric check, with stricter thresholds (60 matches, 40 inliers, ratio 0.50). The edge score for accepted pairs is:

$$\text{score}(i, j) = n_{\text{inliers}} \times r_{\text{inliers}} \times \log(1 + n_{\text{good}}) \quad (3.5)$$

3.6 GRAPH CONSTRUCTION AND CENTRAL IMAGE SELECTION

Accepted pairs form a weighted graph $\mathcal{G} = (V, E)$. Edge pruning retains at most $m = 3$ edges per node. Connected components define the final groups. The central image per group maximises average neighbour score:

$$I_j^* = \arg \max_{I_i \in G_j} \frac{1}{|\mathcal{N}(I_i)|} \sum_{I_l \in \mathcal{N}(I_i)} \text{score}(i, l) \quad (3.6)$$

CHAPTER 4

EXPERIMENTS

4.1 DATASET

Our dataset contains **714 images** from Flickr Commons. After DBSCAN geo-clustering, 364 images form **164 geo-clusters**; the remaining 350 are geographic noise and excluded. Of the 164 clusters, **101 are singletons**, leaving **63 non-trivial clusters** as the actual workload. The largest cluster has 67 images; mean size over non-singleton clusters is 2.2. Table 4.1 shows the full distribution.

TABLE 4.1
Distribution of geo-cluster sizes.

Cluster size	Count	% of clusters
1	101	61.6%
2	32	19.5%
3	17	10.4%
4	5	3.1%
5	4	2.4%
6–8	3	1.8%
20	1	0.6%
67	1	0.6%
Total	164	100%

4.2 EVALUATION PROTOCOL

No manually annotated ground truth is available. We evaluate along two axes: workload reduction (CLIP pre-filtering reduces $\binom{67}{2} = 2211$ pairs to 467, a **79%** reduction) and a configuration comparison across three settings. A sample of 16 groups was also inspected manually via Flickr URLs.

4.3 RESULTS

TABLE 4.2

Pipeline results. All runs use $k = 10$, SIFT + RootSIFT, dual RANSAC. LG = LightGlue + SuperPoint.

Configuration	Processed	Groups	Imgs grouped	Avg size
SIFT + LG, $\tau = 0.75$	32 / 63	11	23	2.09
SIFT + LG, $\tau = 0.55$	61 / 63	9	21	2.33
SIFT only, $\tau = 0.55$	61 / 63	16	37	2.31

Of the 63 non-trivial clusters, 32 produced at least one CLIP candidate pair at $\tau = 0.75$; the remaining 31 had maximum pairwise similarity below τ and produced no pairs, which is expected for geographically proximate but visually dissimilar images. Relaxing to $\tau = 0.55$ raises coverage to 61/63 but yields fewer confirmed groups (9), suggesting additional pairs are geometrically inconsistent. Runtime: 424 s total, 0.55 s per pair on a single CPU core.

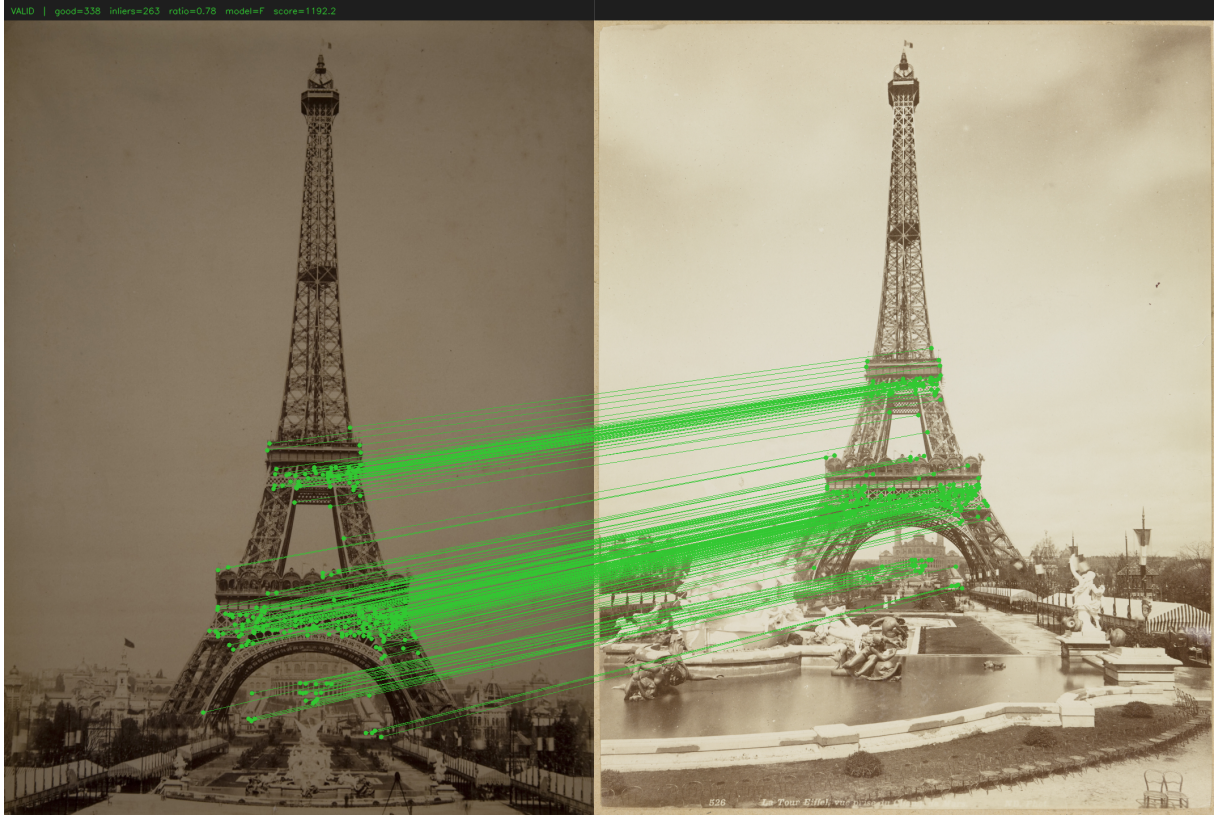


FIGURE 4.1

SIFT + RootSIFT inlier correspondences between two archival Eiffel Tower photographs (good = 338, inliers = 263, ratio = 0.78, score = 1192). Correspondences concentrate on stable structural elements; the fine lattice near the top produces fewer keypoints due to texture ambiguity.



FIGURE 4.2

Discovered group: National Archives, Washington D.C. Four images spanning different seasons, decades, and framings, correctly identified as co-visible observations of the same facade.

CHAPTER 5

DISCUSSION AND CONCLUSION

5.1 DISCUSSION

CURATION OVER RECONSTRUCTION. The average group size of 2.09 images shows that full SfM reconstruction is not the right target here. The pipeline excels at data pruning: it reduces 23 grouped images to 11 central representatives and isolates 341 singletons that share no geometric content with any neighbour.

COVERAGE. Of the 63 non-trivial clusters, 32 yielded at least one CLIP candidate pair at $\tau = 0.75$. The remaining 31 had maximum pairwise CLIP similarity below τ , an expected outcome for geographically proximate but visually dissimilar images.

ABSENCE OF GROUND TRUTH. Without annotated labels, standard precision/recall cannot be reported. Visual inspection of 16 groups suggests qualitatively correct groupings. Labelling 20–30 clusters would suffice for a reliable precision/recall estimate.

CLIP THRESHOLD TRADE-OFF. Lowering τ from 0.75 to 0.55 raises coverage from 32 to 61 clusters but yields fewer confirmed groups (9 vs. 11), confirming that extra candidates are geometrically inconsistent. Higher thresholds may miss co-visible pairs taken decades apart under very different photographic conditions.

LIGHTGLUE AS A PRECISION FILTER. SIFT + LG produces fewer but likely more reliable groups than SIFT alone (11 vs. 16). The additional groups without LightGlue warrant qualitative inspection to assess whether they are true or false positives.

SCALABILITY. At 0.55 s per pair on a single CPU core, the pipeline is tractable for moderate-sized archives. Processing the full Flickr Commons would require a distributed or GPU-accelerated implementation.

5.2 CONCLUSION

We have presented a geometric grouping pipeline for Flickr Commons archival images, designed for intelligent curation rather than 3D reconstruction. By combining CLIP-based candidate pre-filtering, SIFT + RootSIFT local feature matching with CLAHE preprocessing and border masking, dual RANSAC

geometric verification, and LightGlue + SuperPoint symmetric verification, the pipeline partitions noisy geo-clusters into geometrically consistent groups and selects a stable central image per group. On 714 images across 164 geo-clusters, our most selective configuration discovers 11 multi-image groups from 32 processed clusters (average size 2.09), while reducing pairwise matching workload by up to 79%.

Future work should focus on two directions: producing a manually annotated ground truth for quantitative evaluation, and exploring higher-capacity or fine-tuned embeddings to improve recall on domain-specific historical imagery.

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